

Homework 1

Handed Out: Tue September 15, 8:00 p.m. ET Due: Tue September 29, 7:59 p.m. ET

- Submit written HW1 as one legible PDF through Gradescope. You may write your answers in $\text{\LaTeX}$ or scan handwritten work. We cannot award credit for work that cannot be read.
- Assign the correct PDF pages to every question and sub-question in Gradescope. Unassigned pages may receive no credit.
- The submission deadline is **7:59 p.m. ET**. The late penalty is 0.5% of the assignment's maximum score per late hour (rounded up), for at most 48 late hours. Written and coding submissions are timed separately.
- The written and coding halves are graded separately. Written HW1 contains 62 points for CIS 5190 students. The 5-point part marked [**CIS 5190 only**] is optional for CIS 4190 students, whose written maximum is therefore 57 points. Optional work cannot raise a score above 100%.
- Follow the [Human and AI Assistance in Homework policy](#). Do not share or discuss solutions with other students. AI use is permitted only after a documented solo attempt and must be disclosed below.
- If using $\text{\LaTeX}$, copy the [shared solutions template](#). The link is read-only; make a *private* copy before editing.
- Post clarification questions on Ed Discussion. Contact the course staff promptly if you experience submission problems.

1 Use of AI Tools

This section concerns **this written homework**; report coding-homework AI use separately in the notebook. It is graded for being answered, not for any particular answer. Reporting no AI use earns full credit.

1. (2 pts) Did you use an AI tool (for example, ChatGPT, Claude, Copilot, Gemini, or Cursor) on this written homework? If no, say so and you are done with this section. If yes, state:
 - which tool you used and on which question(s);
 - what solo attempt you made before using it;
 - how you used it (for example, checking your derivation, requesting an explanation, or drafting material that you then verified); and
 - how you think its use affected your learning, including if it allowed you to skip understanding something you otherwise would have worked out.

2 Time Spent

1. (2 pts) Report approximately how many hours you spent reading the assignment, reviewing notes, and working on each of Questions 1–5. Decimals are fine (0.5 means 30 minutes); enter 0 for a question you skipped. Fill this in as you go rather than reconstructing it at the end. This section is graded for being answered, not for the amount of time reported.

3 Written Questions

You do not need to show work for multiple-choice items unless a question says otherwise. For derivations and calculation questions, show enough work for the grader to follow your reasoning; an unsupported final answer may not receive full credit.

1. **[Bias–Variance Tradeoff]** (15 pts)

Suppose an ℓ_2 -regularized linear-regression model is trained to convergence with loss

$$\mathcal{L}(\boldsymbol{\beta}; Z) = \frac{1}{n} \sum_{i=1}^n (f_{\boldsymbol{\beta}}(\mathbf{x}_i) - y_i)^2 + \lambda \|\boldsymbol{\beta}\|_2^2.$$

For each change below, assuming it is the only change, state whether it tends to increase, decrease, or leave unchanged the **bias** and **variance** of the regression procedure. A short explanation is required and a partially correct explanation will earn partial credit.

- (a) (2 pts) Increase the number of training examples n .
 - (b) (2 pts) Decrease the regularization parameter λ .
 - (c) (2 pts) Decrease the feature dimension d , where $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^d$.
 - (d) (2 pts) Replace every feature vector $\boldsymbol{\phi}(\mathbf{x})$ by $c\boldsymbol{\phi}(\mathbf{x})$ for fixed $c > 0$, assuming $\lambda = 0$.
 - (e) (2 pts) Increase the gradient-descent step size α , while keeping it small enough that training still converges to the same minimizer.
 - (f) (5 pts) Your fitted model has low training loss and high test loss. Under the assumptions in parts (a)–(e), for each of n , λ , d , c , and α , state whether you should increase it, decrease it, or expect the change to have no effect on test loss.
2. **[Regularization and Sparsity]** (7 pts)

For a parameter vector $\boldsymbol{\beta} \in \mathbb{R}^d$, define the ℓ_1 -regularized linear-regression objective

$$\mathcal{L}_{\ell_1}(\boldsymbol{\beta}; Z) = \frac{1}{n} \sum_{i=1}^n (y_i - \boldsymbol{\beta}^\top \mathbf{x}_i)^2 + \lambda \sum_{k=1}^d |\beta_k|. \quad (1)$$

- (a) (3 pts) Let $R_1(\boldsymbol{\beta}) = \lambda \sum_{k=1}^d |\beta_k|$ be the l_1 regularization term and $R_2(\boldsymbol{\beta}) = \lambda \sum_{k=1}^d \beta_k^2$ the l_2 regularization term. Compute $\partial R_1 / \partial \beta_j$ for $\beta_j \neq 0$ and $\partial R_2 / \partial \beta_j$ for every $j \in \{1, \dots, d\}$. What changes for R_1 at $\beta_j = 0$?
- (b) (2 pts) Does ℓ_2 regularization generally make coefficients exactly zero? Briefly justify your answer using the derivative from part (a).
- (c) (2 pts) Explain why ℓ_1 regularization can produce an optimum with $\beta_j = 0$. Compare its behavior near zero with that of ℓ_2 regularization.
3. **[Closed-Form Linear Regression]** (17 pts)

Let $Z = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, with data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and label vector $\mathbf{y} \in \mathbb{R}^n$. The mean-squared-error objective is

$$\mathcal{L}(\boldsymbol{\beta}; Z) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

- (a) (5 pts) Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, derive the closed-form minimizer $\hat{\boldsymbol{\beta}}$.
- (b) (5 pts) Now minimize $\mathcal{L}_{\text{new}}(\boldsymbol{\beta}; Z) = \mathcal{L}(\boldsymbol{\beta}; Z) + \lambda \|\boldsymbol{\beta}\|_2^2$ for $\lambda > 0$. Derive its closed-form minimizer.
- (c) (2 pts) Is unregularized linear regression guaranteed to have a unique parameter minimizer for every dataset? State the precise condition for uniqueness.
- (d) **[CIS 5190 only]** (5 pts) In simple linear regression with an intercept, $\hat{y}_i = \beta_0 + \beta_1 x_i$. Show that the optimal parameters (β_0^*, β_1^*) satisfy

$$\frac{1}{n} \sum_{i=1}^n (y_i - \beta_0^* - \beta_1^* x_i)(x_i - \bar{x}) = 0, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

4. **[Batch Gradient Descent]** (7 pts)

Consider $n = 2$ examples

$$(\mathbf{x}_1, y_1) = ([1, -1]^\top, 0), \quad (\mathbf{x}_2, y_2) = ([-1, -1]^\top, 1),$$

and the ℓ_2 -regularized linear model $f_{\boldsymbol{\beta}}(\mathbf{x}) = \boldsymbol{\beta}^\top \mathbf{x}$ trained with loss function

$$\mathcal{L}(\boldsymbol{\beta}; Z) = \frac{1}{n} \sum_{i=1}^n (y_i - \boldsymbol{\beta}^\top \mathbf{x}_i)^2 + \lambda \|\boldsymbol{\beta}\|_2^2.$$

Initialize $\boldsymbol{\beta}^{(0)} = [0, 0]^\top$ and use **full-batch gradient** descent with $\lambda = 1$ and step size $\alpha = 1$.

- (a) (2 pts) Compute $\mathcal{L}(\boldsymbol{\beta}^{(0)}; Z)$, the loss at $\boldsymbol{\beta} = \boldsymbol{\beta}^{(0)}$.
- (b) (5 pts) Derive the gradient, compute $\boldsymbol{\beta}^{(1)}$ and $\boldsymbol{\beta}^{(2)}$, and report $\mathcal{L}(\boldsymbol{\beta}^{(2)}; Z)$. Show your work.

5. [Diagnosing Gradient Descent] (4 pts)

Consider the following separate scenarios for a linear regression model being trained with gradient descent:

- (a) (2 pts) The training loss oscillates and sometimes increases. What is the likely cause, and how would you address it?
- (b) (2 pts) After 1,000 iterations, the training loss is still decreasing at a steady but slow rate. What might this indicate, and what would you check or change?

6. [Covariance as a Linear Operator] (8 pts)

We are used to thinking of eigenvectors as the directions that are left unchanged by a linear transformation. In this question you will understand covariance Σ as a linear operator, and then see how to interpret its eigenvectors and eigenvalues. Throughout, let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ be a **centered and scaled** dataset and use the fact that $\Sigma = \frac{1}{n} X^\top X$ can also be written as (not part of the graded HW, but if you don't see this right away, work it out and convince yourself that these are equivalent!):

$$\Sigma = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top, \quad (2)$$

- (a) (3 pts) Starting from Equation 2, show that for any “probe” vector $\mathbf{v} \in \mathbb{R}^d$,

$$\Sigma \mathbf{v} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^\top \mathbf{v}) \mathbf{x}_i. \quad (3)$$

[Hint: this is one line. All you need is that $\mathbf{x}_i^\top \mathbf{v}$ is a scalar, so it can be moved freely through any product terms.] Equation 3 writes $\Sigma \mathbf{v}$ as a weighted sum of the data points themselves: each data point “pulls” the vector summation in the RHS along its own direction, with weight proportional to $\mathbf{x}_i^\top \mathbf{v}$. In one sentence, using a geometric interpretation of dot products, what does this weight on $\mathbf{x}_i$ measure? When is that weight zero, and what does that mean geometrically?

After working this out, you should be able to appreciate the geometric nature of the linear transformation applied by Σ to various vectors in $\mathbb{R}^d$.

- (b) (2 pts) For the rest of this question use the $n = 4$ mean-centered samples ($d = 2$)

$$\mathbf{x}_1 = (2, 2)^\top, \quad \mathbf{x}_2 = (-2, -2)^\top, \quad \mathbf{x}_3 = (1, -1)^\top, \quad \mathbf{x}_4 = (-1, 1)^\top.$$

Verify that $\Sigma = \begin{pmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{pmatrix}$. Then compute $\Sigma \mathbf{v}$ with $\mathbf{v} = (1, 0)^\top$: using Equation 3 rather than matrix multiplication, tabulate the four weights $\mathbf{x}_i^\top \mathbf{v}$ and the four contributions $\frac{1}{4}(\mathbf{x}_i^\top \mathbf{v})\mathbf{x}_i$, and add them up. Observe how each data point applies a weighted “pull” on the outcome along its own direction, as noted above in part (a).

- (c) (2 pts) Now probe the same dataset with $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)^\top$. Two of the four weights are zero; which samples, and why? Show that the two surviving contributions both point along $\mathbf{u}$ itself, and conclude that $\Sigma\mathbf{u} = 4\mathbf{u}$.
- (d) (1 pt) Based on the above exercises, complete the following sentence in your own words, then justify it: “ $\Sigma\mathbf{v}$ is parallel to $\mathbf{v}$ exactly when ...” Why does this make the major axis of a data cloud an eigenvector of Σ ?